Usability testing

Lecture 7, IT Product Development

Innopolis University, 2026

Agenda

- Usability testing
- Running the test
- Post-test surveys
- Analyzing the results

Usability testing

Review

What is usability?

Why do you think you need to test it?

When you were building your app, did you think about affordances?

What are Gibson affordances? (Exploratory actions)

What are Norman affordances? (Correct usability)

Usability testing

Action:

- Evaluate a product or service by testing it with representative users.

Goal:

- Identify any usability problems.

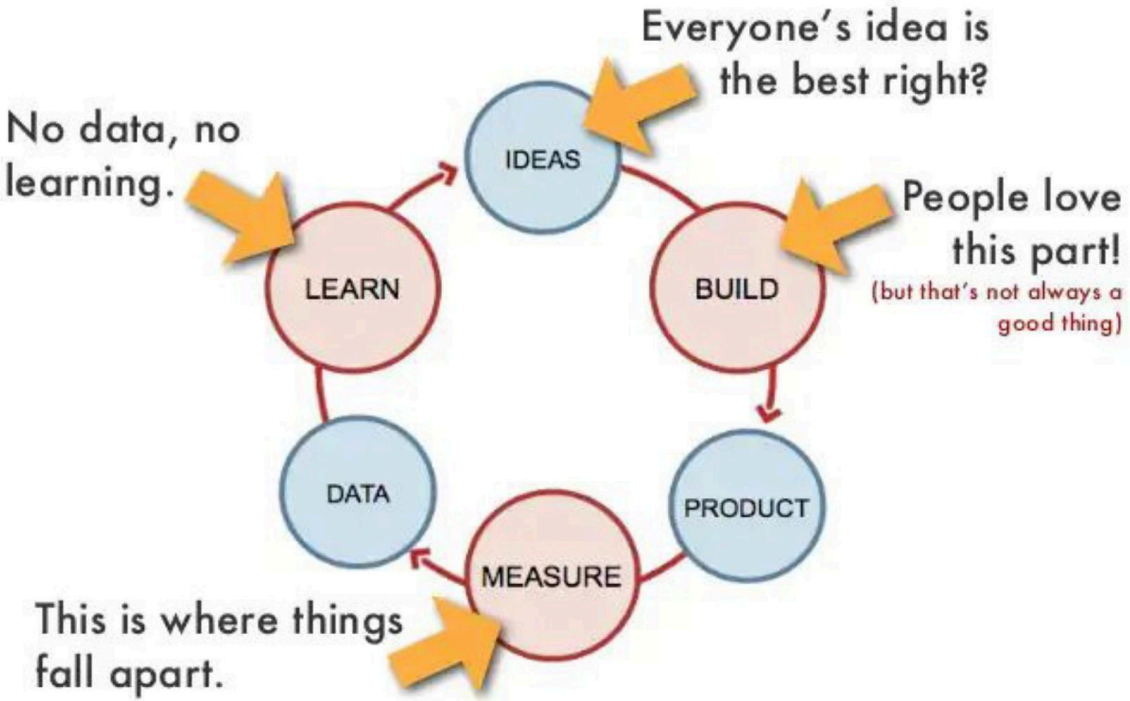
How:

- Collect qualitative and quantitative data.
- Determine the participant’s satisfaction with the product.

Usability testing: focus

Focus on:

- Measure
- Data
- Learn



Usability testing process outline

1. Run pre-test survey
2. Run experiment (test)
3. Collect data
4. Run post-test survey
5. Organize
6. Interpret and analyze
7. Learn

Running the test

Pre-test survey

The goal is to understand the person and compare results between users with similar backgrounds after the tests.

A basic “Please, tell us about yourself” will do, but try to get more specific details that matter for your application.

Run experiment

Give users an ability to use the system without interruptions.

No guidance, help, or user manual is given to the user.

Collect data

It is best if you have automatic action data collection.

You can observe what the user does and take notes.

Post-test surveys

Post-test survey

Survey frameworks:

- Single Ease Question (SEQ)
- After-Scenario Questionnaire (ASQ)
- System Usability Scale (SUS)
- Open-ended questions

Single Ease Question (SEQ)

Overall, how difficult or easy was the task to complete?



After-Scenario Questionnaire (ASQ)

1. Overall, I am satisfied with the ease of completing the task in this scenario.
2. Overall, I am satisfied with the amount of time it took to complete the task in this scenario.
3. Overall, I am satisfied with the support information (online help, messages, documentation) when completing the task.



System Usability Scale (SUS)

Standard 10 questions in order:

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.

…

With the following response format:

Strongly Disagree 1	2	3	4	Strongly Agree 5
○	○	○	○	○

Source and details: https://measuringu.com/sus/

Open-ended questions

The goal is to fill in missing context.

When preparing questions, use the approaches that we discussed:

- ORID
- “The Mom Test”

Analyzing the results

Organize: observation

Exploratory actions / Gibson affordances	Correct usability / Norman affordances
The user was using the Ctrl-F key to get to the “search” feature	The user saw the search bar on the interface and used it
…	…

Organize: match user

Match each user to the corresponding:

- pre-test survey
- specific experiment
- observation
- post-test survey

Interpret and analyze: example

Interpreting SUS results:

By adding the scores together you get a 0-40 scale.

To convert it to 0-100, multiply by 2.5.

The result **is not a percentage!**

Interpret and analyze: example

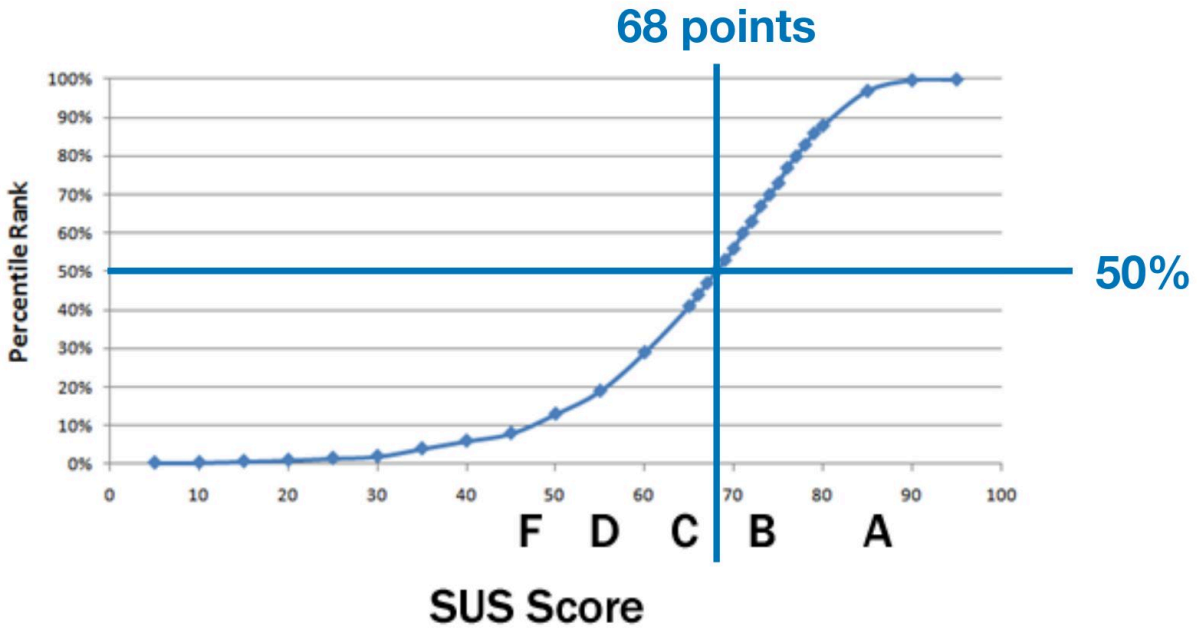
Interpreting SUS results:

The average score, based on 500+ case studies, is

68 points = 50 percentile

A score below 68 is considered below average, or below 50%, and vice versa.

Interpret and analyze: example



Learn

Write down what you have learned after interpreting and analyzing the results.

Review

What is usability?

Why do you think you need to test it?

What do we focus on in usability testing?

Describe the usability testing process.

Describe the surveys that we can use.

What is SUS?

How do we interpret SUS results?

Questions?